Paleomagnetic and Rock-Magnetic Studies of the Permian Cutler and Elephant Canyon Formations in Utah[1]

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The Permian Cutler formation and the upper 15 meters of the Permian Elephant Canyon formation have been collected at 0.6-meter stratigraphic intervals southwest of Moab in eastern Utah. The NRM directions show a pronounced streak distribution, but thermal demagnetization successfully isolates the stable paleomagnetic direction. All directions are reversed and no significant long-term change in pole position is observed throughout the entire section. The pole calculated from the Elephant Canyon data lies at 43.6°N, 119.6°E; the Cutler pole lies at 44.4°N, 116.2°E. Rock-magnetic analyses suggest that the secondary magnetization results from the iron hydroxides and was acquired after recent surface exposure.

In recent years several authors have called attention to the possibility of using magnetic reversals as a tool for stratigraphic correlation [*Picard*, 1964; *Burek*, 1967, 1970; *McMahon and Strangway*, 1968*a*, *b*; *Helsley*, 1969]. Before this can be done successfully on a worldwide scale a standard section must be established. This requires dense stratigraphic sampling where the spacing depends on parameters such as rate of deposition and continuity of the section. For example, the lowest normal interval in the Moenkopi formation extends stratigraphically over 18 meters at Gateway, Colorado, and thins to only 3 meters in Monument Valley, Arizona [*Helsley*, 1969]. The thinning can be explained by a decrease in the sedimentation rate away from the source area. In this study, 353 samples were collected by means of a portable drill [*Helsley*, 1967] at an average spacing of approximately 0.6 meter. The Permian Cutler and Elephant Canyon formations were chosen because of the importance of the Kiaman magnetic division for worldwide stratigraphic correlation.

The Cutler formation extends through most of southeastern Utah and southwestern Colorado. It is of highly variable lithology. Toward the Uncompaghre plateau, one of the source areas [*Baker*, 1946], the Cutler formation consists of arkose and conglomeratic arkose [*Baars*, 1962] and is considered to be of fluvial origin. The samples for the present study come from an area southwest of Moab, Utah (38°25′N, 109°35′W). Here the Cutler formation consists of massive sandstone interlayered with siltstone and occasional conglomeratic lenses. The sandstone is reddish brown in color, medium-fine grained and well cemented. Although the Cutler formation has not been subdivided into members at this locality, it can be subdivided into 5 members a few kilometers to the west. These are the Halgaito shale (which interfingers with the Elephant Canyon formation), the Cedar Mesa sandstone, the Organ Rock shale, the White Rim sandstone, and an unnamed shale that overlies the White Rim sandstone. In the area sampled the upper portion of this section is similar in lithology to the Organ Rock shale of the Monument Valley region and probably includes its equivalents as well as the unnamed shale member that overlies the White Rim sandstone. Toward the bottom of the section a massive tan colored sandstone was encountered that may be the equivalent of the Cedar Mesa sandstone member. The lowest stratigraphic interval is rather similar to the upper section and probably represents portions of the Halgaito shale member. The total thickness of the Cutler formation at this locality is about 200 meters and the beds dip 2° to the south.

The Cutler formation is underlain by a formation formerly referred to as the Rico forma-

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tion or Rico transition. *Baars* [1962] and *Baars and Seager* [1970] have shown that this usage is incorrect and have suggested inclusion of the Pennsylvanian portion of these rocks as a facies of the Honaker Trail formation and assignment of the unconformably overlying Permian (Wolfcampian) portion to the Elephant Canyon formation. This usage is followed in this paper. The Elephant Canyon formation consists of interbedded sandstone, shale, siltstone, and fossiliferous limestone, with sandstone predominating, all of Wolfcampian age. The portion of this formation between 10 and 25 meters below the base of the Cutler formation is included in this study.

Since the Cutler formation does not contain any fossils at this locality its age can only be inferred from the regional stratigraphy. It is conformably underlain by, and at some localities interfingers with, the fossiliferous Elephant Canyon formation. *Baars* [1962] has shown the Elephant Canyon formation to be Wolfcampian (Lower Permian) in age. At the collection site the Cutler formation is overlain with apparent conformity by the Hoskinnini member of the Moenkopi formation, which is considered to be Lower (?) Triassic age [*Stewart,* 1959]. Regionally, however, an angular unconformity is observed between the Cutler formation and the Hoskinnini member; this pattern suggests that even at this site a considerable portion of the Permian section is missing. The uppermost shales sampled are probably equivalent to the shale overlying the White Rim sandstone that is believed to be of early Leonardian age [*Baars and Seager,* 1970]. Consequently the portion of the Cutler sampled in this study probably ranges from middle Wolfcampian to middle Leonardian in age, a range encompassing most of the Lower Permian.

The Cutler formation was first studied paleomagnetically by *Graham* [1955] in a survey paper on Paleozoic and Mesozoic rocks from the western United States. More recently *Farrel and May* [1969] collected the entire Cutler formation in the Monument Valley area but had to reject 90% of their data because of magnetic instability. Of their samples only those from the Halgaito shale, the basal member of the Cutler formation, gave reliable results. *Helsley* [1971] determined a pole position from the arkosic sandstones of the Cutler formation of western Colorado and suggests a 20° change in pole position between the Lower Permian and Lower Triassic.

Other studies of Permian rocks in the region surrounding the Colorado plateau have been made by *Runcorn* [1955, 1956], *Doell* [1955], and *Collinson and Runcorn* [1960]. The pole positions derived from these data are no longer considered reliable because no stability experiments were performed. *McMahon and Strangway* [1968*a*, *b*] investigated the red beds of the Colorado front range in order to define the extent of the Kiaman magnetic division. The average spacing of their samples is 30 meters, and short normal events, if they exist, could easily have been missed. The present study was initiated to determine whether the Kiaman division is indeed predominantly of reversed polarity and to test the amount of polar wandering during the part of the Permian represented by the Cutler formation.

MAGNETIC PROPERTIES

A few samples representing all lithologies encountered were examined microscopically. In all cases, hematite occurs in two modes, granular and interstitial. The hematite grains usually contain magnetite relics (martite). Whether this reflects a highly oxidized source material or whether the hematite formed from magnetite during subsequent diagenesis cannot be decided. The interstitial hematite is observed as a film around quartz grains and in the cement. The only other magnetic mineral is goethite, which is present as small patches and occasionally as a surface film.

Twenty-six samples were used for stepwise thermal demagnetization and Curie point determinations in order to differentiate between the magnetic minerals that are present and those that carry a magnetic remanence. The Curie points were measured with a magnetic force balance using original, nonenriched powder samples.

The highly variable response to thermal demagnetization can be bracketed by the two examples shown in Figure 1. Figure 1*a* shows a primarily one-component system, namely hematite. The slight initial increase in intensity can be explained by the removal of a small component aligned along the present field. The

directions remain rather constant; most of the change occurs below 150°C. The other extreme is demonstrated in Figure 1*b*. The intensity drops markedly upon heating to 150°C. This decrease is associated with pronounced systematic changes in direction. Although the intensity does not decrease much above this temperature, the directions continue to approach the Permian direction up to about 400°C. Above this temperature no changes larger than about 10° occurred in any sample. Of these samples, 13 decreased in intensity to less than 60% of their NRM (natural remanent magnetization) value upon heating to 200°C. From this experiment we can conclude that two magnetic phases carry a remanence; one is removed by heating to about 200°C, and the other one is stable up to at least 600°C.

The Curie point curves (Figure 2) exhibit several interesting features. The gradual decrease in the first example (Figure 2*a*) can be explained by the presence of hematite with a very broad blocking temperature spectrum [*Néel*, 1955]. Superimposed on this we find a trace of magnetite. The increase of the magnetization at 350°C coincides with the transition of goethite to hematite. The cooling curve and repeated heating of this sample show that all magnetic minerals had converted to hematite. Results identical to those in Figure 2*a* were obtained, however, when previously unheated material was used. The second example (Figure 2*b*), which is typical of most samples, first increases in intensity and then drops gradually, showing a weak magnetite Curie point and finally a hematite Curie point.

After cooling to room temperature, the intensity regains only two-thirds of its initial value. The behavior of this sample strongly suggests the presence of small amounts of lepidocrocite. The increase in intensity upon heating is interpreted as reflecting the dehydration of lepido-

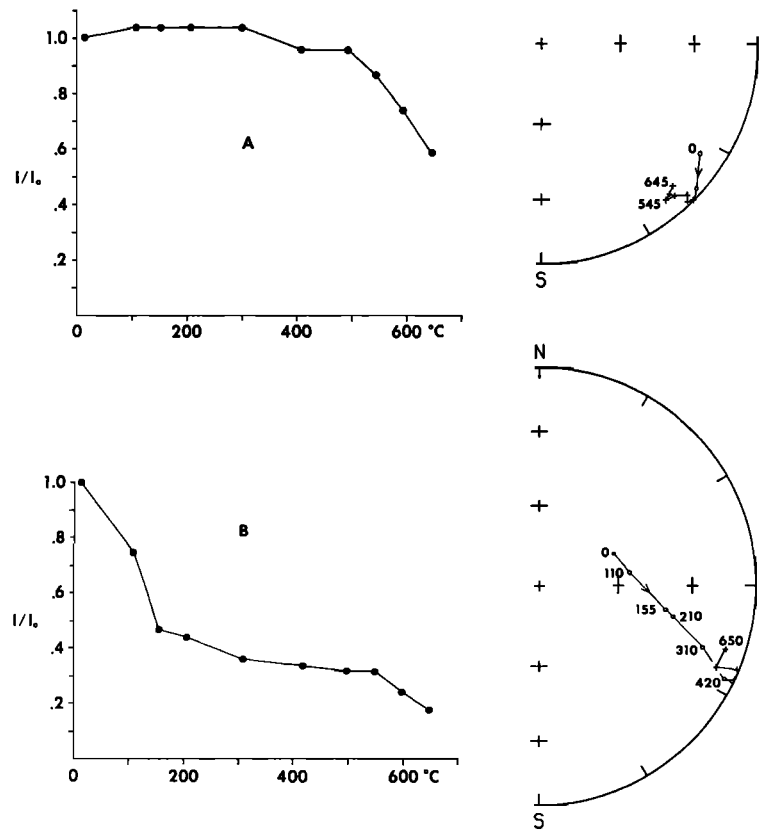


Fig. 1. Response to thermal demagnetization. Arrows indiate the direction of change with increasing temperature.

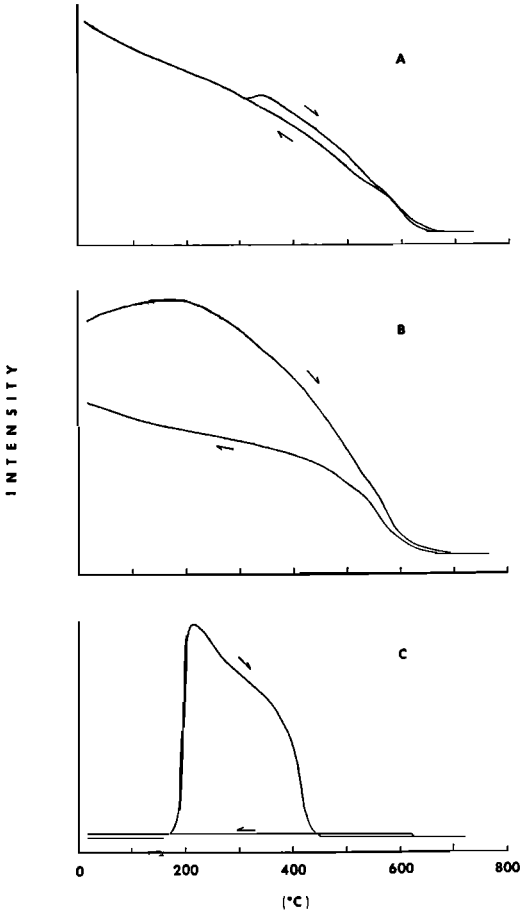


Fig. 2. Thermomagnetic variation of two rock samples (*a* and *b*) and synthetic lepidocrocite (*c*). Arrows indicate direction of change. Vertical scale is arbitrary.

crocite to maghemite ($\gamma - Fe_2O_3$). Upon further heating maghemite converts to hematite ($\alpha - Fe_2O_3$). The fact that the intensity after heating is lower than before heating indicates that this sample may have contained some maghemite in its original state and it is considered that it was formed from the iron hydroxide. In order to substantiate this interpretation, a Curie point curve of synthetic lepidocrocite is included as Figure 2*c*. It may be added that we verified its purity and, in particular, its nonmagnetic character with the Mössbauer effect.

The magnetic properties of the Cutler formation in this section are remarkably uniform in spite of substantial lithological variations. All tested samples show a broad hematite blocking temperature spectrum. In addition, the Curie point measurements point out the importance of iron hydroxides in sediments as a contributor to their NRM properties. Although only goethite was identified microscopically, the Curie point determinations strongly suggest the presence of lepidocrocite as well. Thus, the main difference in the NRM behavior between individual samples is thought to be the variation in the iron-hydroxide content.

PALEOMAGNETIC DATA

Cutler formation. The directions of the NRM are distributed along the great circle joining the expected Permian with the present day direction (Figure 3). Since the axial dipole direction ($D = 0°$, $I = 58°$), the direction observed today ($D = 15°$E, $I = 65°$), and the Permian direction ($D = 160°$, $I = -16°$) lie on the same great circle, we cannot use this distribution to obtain information about the time constant of the viscous component. Only a few data points lie far removed from this streak and no sample has a definite normal direction. If the NRM directions are rotated in such a way that all points lie within one hemisphere, it becomes quite apparent that the distribution widens by a factor of two or so as the directions approach the present day direction (for a discussion see section 3).

One specimen of each sample has been thermally demagnetized. The furnace was located inside a moly-permalloy room and the small residual field was cancelled to within ±3 gammas by two pairs of Helmholtz coils. A demagnetizing temperature of 500° to 540°C was chosen on the basis of previous laboratory experience. The appropriateness of this choice is substantiated by the observations made on the stepwise demagnetized samples in which only small and random changes in direction were observed above 540°C.

The grouping of the directions improves considerably after thermal demagnetization (Figure 3). All directions are reversed and less than 5% have directions more than 45° away from the mean. The mean direction coincides with the directions of those NRM data that were initially in the upper hemisphere at the end of the streak farthest from the present field direction and that underwent little or no change during demagnetization (see Figures 3

and 4). This is a good indication that the original direction has been attained. In order to further ensure that no secondary component was left, all specimens whose direction deviated more than 20° from the mean were demagnetized at 620°C as well. A few samples moved closer to the mean, but most samples showed only small erratic changes in direction.

The vectors that were removed during demagnetization, that is, $I_{NRM} - I_{520°C}$, are shown in Figure 4. The distribution of the directions clearly demonstrates that the secondary component is indeed primarily a present day component and does not originate in the distant geological past. About 70% of the samples decreased in their intensities to less than 70% of their initial value; 50% of the samples decreased to less than half their NRM intensities (Figure 4). There is no correlation between intensity decrease and proximity to the mean. This is in contrast to the observations by *Farrel and May* [1969], who argue that all samples that retain less than 80% of their NRM intensity after heating to 500°C are magnetically unstable. This discrepancy can probably be explained by the fact that Farrel and May cooled their samples in the presence of a 200-gamma field. From experience in our laboratory we have learned that even a few tens of gammas are sufficient to add a magnetization to a sample that easily can be of the order of 10^{-5} to 10^{-6} emu/cm^3, an intensity range commonly encountered in red sandstones.

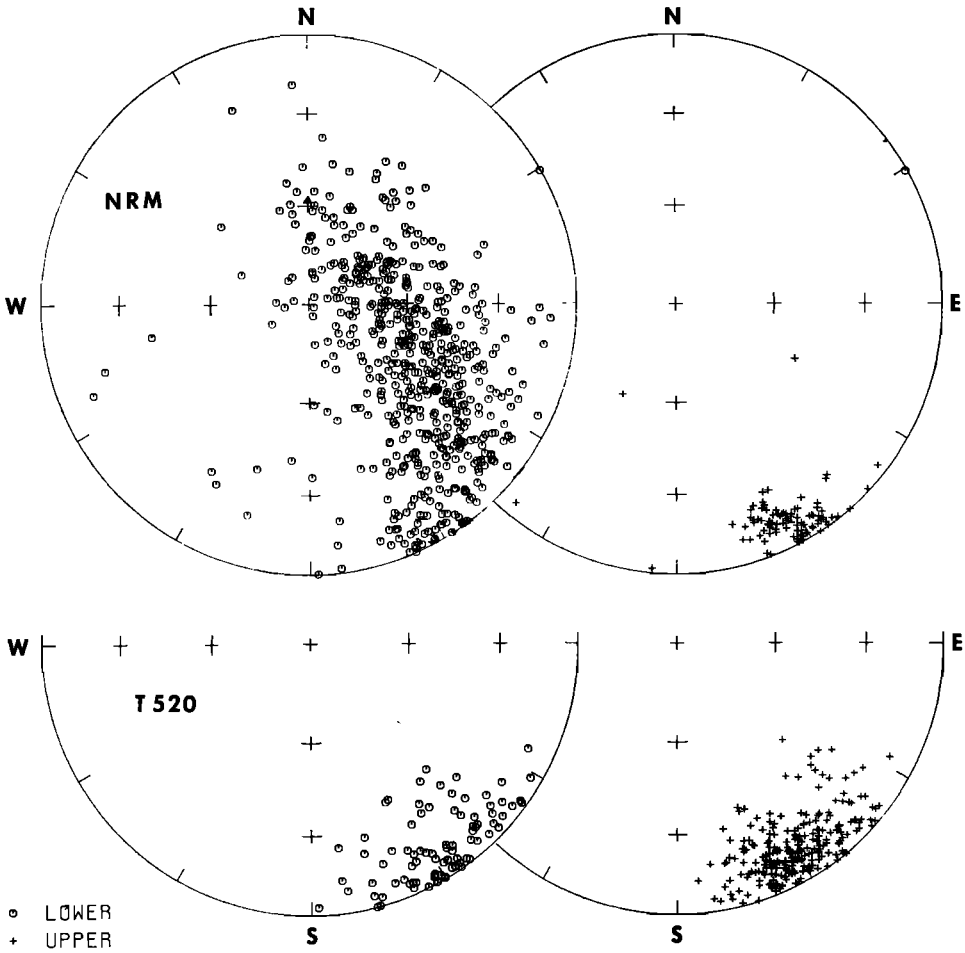


Fig. 3. Direction of magnetization of Cutler samples before and after thermal demagnetization. Left half of figure represents lower hemisphere; right half represents upper hemisphere. Equal area projection. Same symbols apply to all figures.

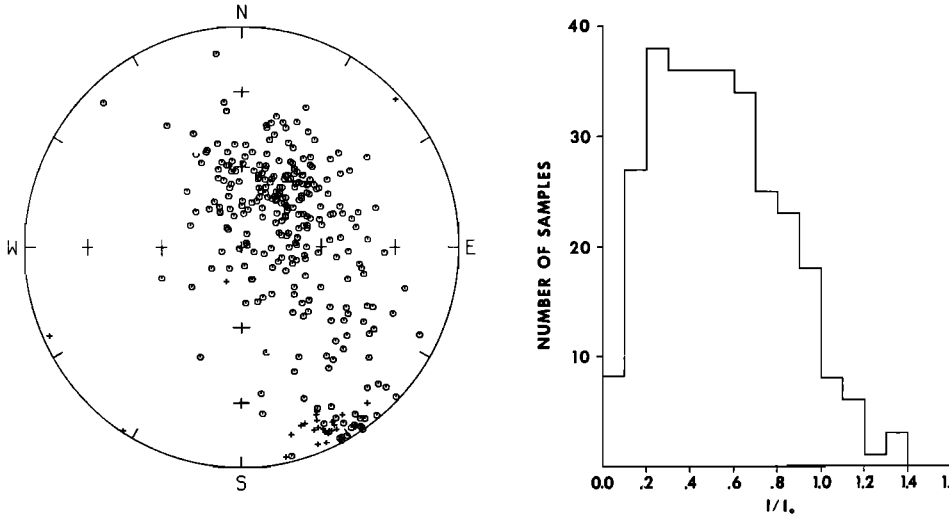


Fig. 4. Direction of magnetization that was removed upon heating to 520° ± 20°C and relative intensity decrease.

Figures 5 and 7 show the intensity and direction of magnetization of all samples as a function of their stratigraphic position. The section between 25 and 50 meters above the base of the Cutler formation was not collected, owing to the lack of suitable outcrops. The intensities of the NRM are approximately 10^{-6} emu/cm^3, and no major variations are observed. The NRM directions show a large scatter throughout the entire section.

After thermal demagnetization the declination and inclination remain essentially constant, suggesting that any change in pole position during Cutler time was at best small. This was statistically tested by fitting a least-squares straight line through the declination and inclination data shown in Figure 5 (columns D(T520) and I(T520)). Throughout this section the change in the inclination is insignificant (1° maximum change) whereas the declination data yield a 5° change between the bottom (D = 147.3°) and top (D = 152.3°) of this profile. However, this change lies just within the error limits of this linear fit and therefore is not statistically significant (compare section on pole positions). It is interesting to note, though, that the changes suggested by this test are such that they approach the direction of the Lower Triassic Moenkopi formation [*Helsley*, 1969].

Because of the dense sampling it was considered worthwhile to test the data for long-term variations of the Permian field with periods of the order of 100,000 to 1,000,000 years. Figure 6 shows the gliding average over 10 samples (10 (TOTAL)) and 20 samples (20 (TOTAL)). The horizontal scale has been highly expanded as compared to Figure 5. Rather systematic variations around the mean are indeed apparent and the behavior for both cases is very similar. In order to check the reality of the variations, the data set was subdivided into two subsets, the odd sample numbers and the even sample numbers. This in effect produces two nearly identical populations from the same time sequence, since all the fluctuations in the 10- and 20-sample averages take place gradually so that the difference between 0.6- and 1.2-meter spacing is insignificant. In order to average over the same stratigraphic (i.e., time) interval the curves labeled 10 (ODD) and 10 (EVEN) are actually gliding averages over 5 samples only; similarly, the curves 20 (ODD) and 20 (EVEN) represent 10-sample averages. Although some discrepancies do exist, the main features clearly repeat in all curves and the variations are therefore considered to be real. Our data imply that the amplitudes of the variations of the Permian geomagnetic field (see also Table 1 and Figure 12) are similar to the present day variations.

Elephant Canyon formation. The Elephant

Canyon formation conformably underlies and laterally interfingers with the Cutler formation in this area and is of Lower Wolfcampian age [*Baars*, 1962]. Thirty-two samples were collected from the 25 meters of section immediately below the Cutler formation. Magnetically these samples behave rather similarly to the Cutler samples. The NRM directions exhibit a great circle distribution which upon heating to 520 ± 20°C becomes a fair cluster in the southeast quadrant (Figure 8). Stratigraphically no gross changes are observed (Figure 7) and the mean direction is primarily the same as that of the overlying Cutler formation.

ORIGIN OF UNSTABLE MAGNETIZATION

The Curie point measurements show that most samples have blocking temperatures throughout the entire temperature range. For the natural remanence, however, two narrow blocking temperature regions seem to be of particular importance: the intervals between 150° and 200°C and between 600° and 650°C. According to the theory of *Néel* [1955] the high temperature component is stable over geological times at normal surface temperatures, whereas the low temperature component has a maximum relaxation time of only 1000 years under normal surface conditions. The low temperature component is therefore a viscous remanent magnetization (VRM) whose direction should be aligned along the present earth's field. Figure 4 clearly shows that this is indeed the case.

We noted earlier that the NRM directions deviate more from the great circle joining the

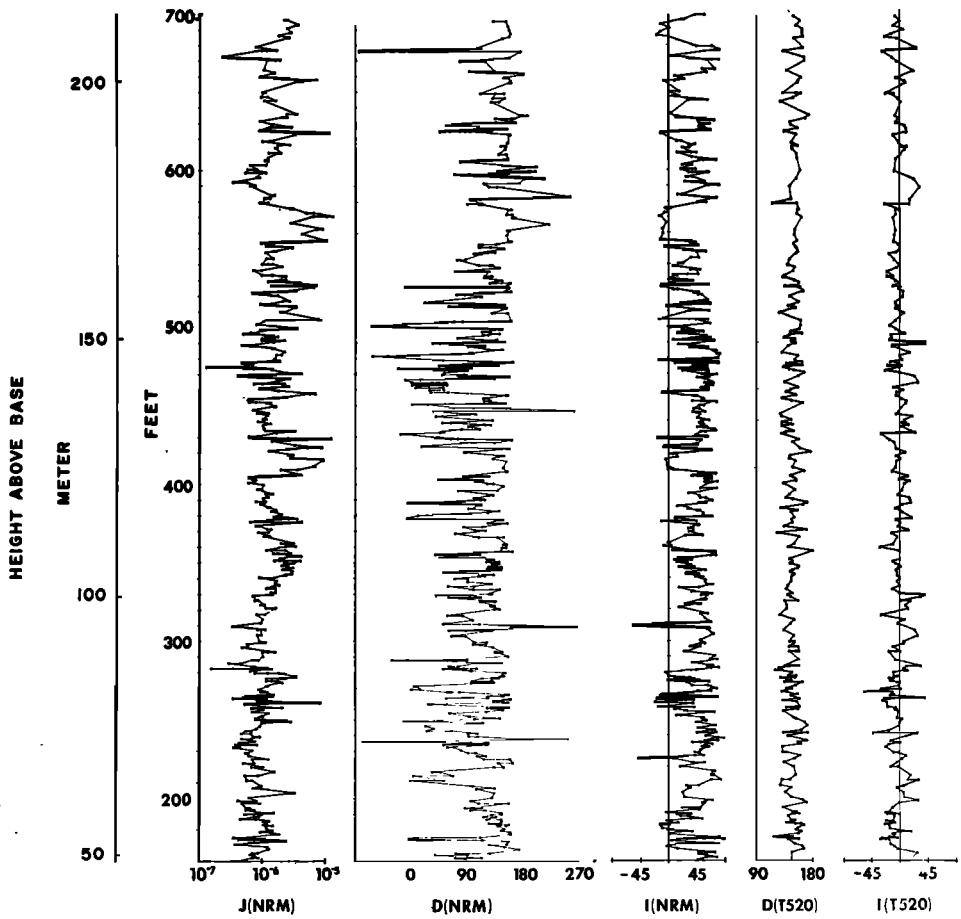


Fig. 5. Stratigraphic distribution of intensity (in emu/cm^3), declination (positive to the east), and inclination (positive downward) of NRM and thermally demagnetized data.

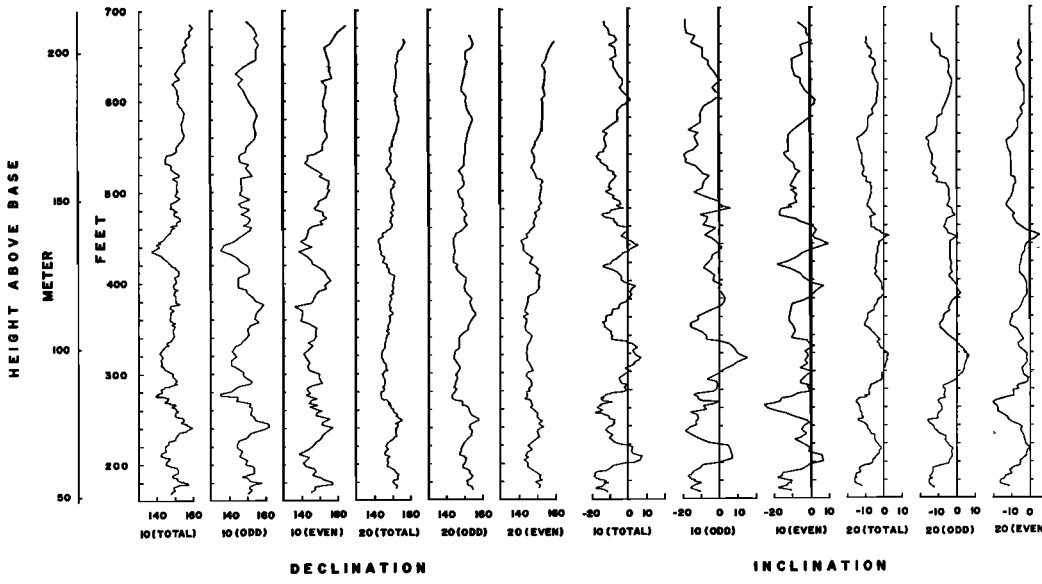


Fig. 6. Declination and inclination of a gliding average over 10 and 20 samples of the thermally demagnetized data shown in Figure 5. TOTAL is the average of all the samples while ODD and EVEN are averages over independent sets of samples consisting of the odd and even numbered field specimens, respectively. Note the marked similarity of the 20 (ODD) and 20 (EVEN) curves.

Permian and present axial direction the farther they lie from the Permian direction. This behavior cannot be explained with only two components. As far as stability is concerned we must assume a three-component system. The first component has a relaxation time long enough to maintain the Permian field direction. The second component is stable over a period of a few years up to 1000 years or more. Superimposed we have a component that has been acquired in the laboratory as is evidenced by changes in direction when repeat measurements are made after several months of storage. This magnetization is apparently more pronounced in those samples for which the secondary component is strong and is absent in those samples that originally were close to the Permian direction.

The mineral responsible for the unstable components can be either hematite with a sufficiently short relaxation time or one of the iron hydroxides. Although hematite cannot be ruled

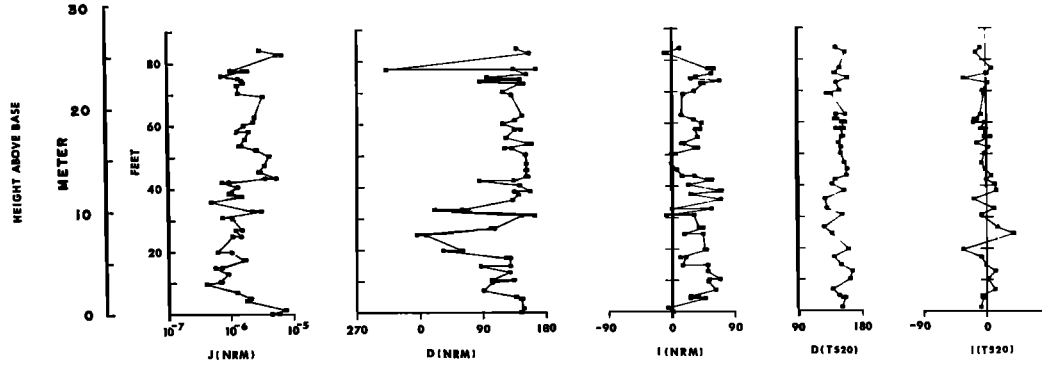


Fig. 7. Intensity, declination, and inclination of basal 25 meters of Cutler formation.

out, it is more likely that the unstable magnetization results from the iron hydroxides. Goethite was observed microscopically and verified with the Curie balance (Figure 2*a*) and is known to possess a stable remanent magnetization [*Strangway et al.*, 1968; *Banerjee*, 1969]. The importance of goethite as the carrier of the secondary component has been demonstrated by *Strangway et al.* [1968]. Surface samples from the Permian Lyons sandstone were magnetically unstable and contained goethite, whereas subsurface samples did not contain goethite and were stable. Similarly, the NRM directions of surface samples from the Triassic Moenkopi formation have a large scatter but equivalent subsurface samples do not [*Helsley*, 1969; C. Baag and C. E. Helsley, unpublished manuscript, 1972]. Although we do not have subsurface samples from the Cutler formation, it is quite possible that our samples also obtained their secondary magnetization after surface exposure, since, as has been discussed by *Walker* [1967], goethite may form instead of hematite under arid conditions.

Strangway et al. [1968] showed that goethite acquires most of its magnetic remanence upon heating to about 200°C and subsequent cooling in the presence of a field although the Néel temperature of goethite is much lower; it lies between 97° [*Hrynkiewicz et al.*, 1965] and 120°C [*Strangway et al.*, 1968]. Four specimens were subjected to this test. They were heated in 25° steps up to 625°C and a 1-oersted field was applied during cooling. The three samples that had a strong secondary component showed a maximum around 200°C (Figure 9*a*), which would be due to goethite according to *Strangway et al.* [1968]. It seems possible, though, that this maximum can be explained in another way. Our Curie point measurements indicate the presence of lepidocrocite which converts to maghemite at about this temperature (Figure 3*b*, *c*). Since the magnetic moment of maghemite is very much larger than that of goethite, only small amounts need be present and may easily escape detection by X-ray defraction. At the moment this intepretation is only tentative because more data are needed for a firm statement.

Figure 9*b* shows the partial thermoremanence of a magnetically stable sample. The fact that there is no indication of a maximum around 200°C supports the idea that the secondary component does indeed result from the presence of iron hydroxides.

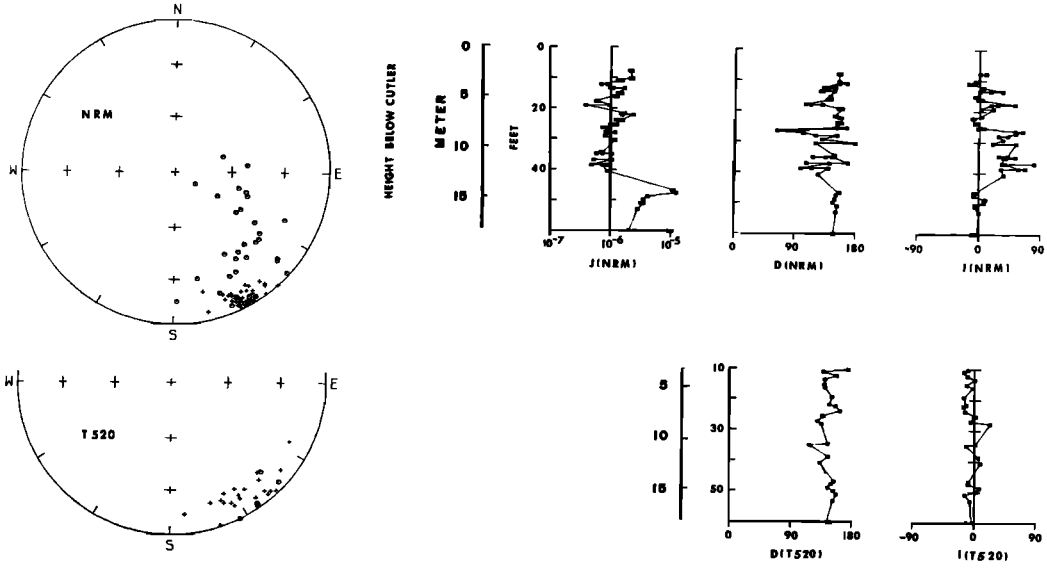


Fig. 8. Stereographic and stratigraphic distribution of Elephant Canyon samples before ($N = 67$) and after ($N = 28$) thermal demagnetization.

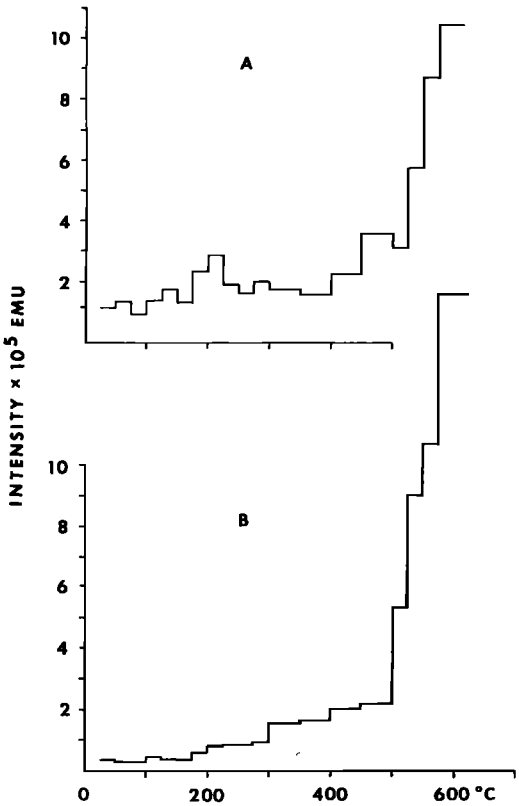


Fig. 9. Partial thermoremanence of unstable (*a*) and stable (*b*) sample.

ORIGIN OF STABLE MAGNETIZATION

Microscopic studies revealed two phases that may be the carrier of the original magnetization, namely the interstitial hematite and the opaque martite grains. The magnetite relics can be ruled out as carriers of a stable remanence because after heating above 580°C the directions remained well grouped and hardly any intensity decrease was observed between 500° and 600°C. In addition, the magnetite within the martite grains amounts to only 10% or so and one therefore would expect that its direction of magnetization is determined by the hematite, owing to exchange interaction [*Porath*, 1968].

In an attempt to discriminate between detrital remanent magnetization (DRM) and chemical remanent magnetization (CRM) seventeen specimens were chemically demagnetized [*Collinson*, 1967; *Burek*, 1969]. In this technique about 1 liter of concentrated hydrochloric acid is forced through a sample by means of compressed air. The sample is then flushed with water, dried, and measured. The degree of bleaching was determined by cutting the sample along two perpendicular planes. Microscopic examination of the samples showed that the opaque grains were hardly affected by the acid.

The results are shown in Figure 10. Two specimens are not included because they were unstable after the treatment. The fact that the large opaque grains remained intact strongly suggests that the stable magnetization does not reside in these grains and thus, if the magnetization is of detrital origin, it is not related to the large grains. In general, the directions move toward the Permian direction; however, they do not cluster nearly as well as the thermally demagnetized samples. This difference seems to indicate that the stable magnetization is to be associated with the least soluble fraction but that it cannot be isolated by chemical means. Incomplete bleaching, the limited number of samples, or original scatter are some of the possible causes of the poor grouping observed. The intensities decrease approximately linearly with the degree of bleaching. The well bleached samples retained less than 20% of their initial value; this phenomenon is to be compared with an average intensity drop of about 50% upon thermal demagnetization (Figure 4), and it suggests that at least part of the stable magnetization resides in the hematite film. Although the experiments are somewhat inconclusive, the large detrital grains are apparently not responsible for the remanence. Thus, all of the remanence could be a CRM or it could be a DRM from the smallest opaque grains combined with the CRM of the cement.

The age of the stable remanence can only be inferred indirectly, since no field test was possible. At the sampling locality the Cutler formation is of fluvial origin and was deposited in a highly oxidizing environment. If the remanence is of detrital origin, then the fluvial environment seems to warrant the conclusion that the acquisition of a remanence coincides with the time of deposition. *Walker* [1967] suggests a similar environment for the Colorado red beds and the recent to Pliocene deposits in Baja California. He shows that the hematite pigment in the Baja California red beds formed in situ over a time period of less than 10 m.y. Moreover, Mössbauer studies prove that the

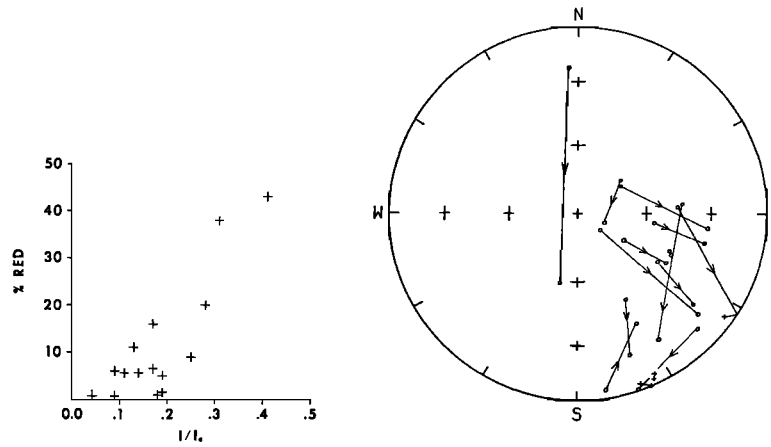


Fig. 10. Change in intensity and direction after chemical demagnetization.

film is magnetic. The fact that hematite occurs in the Cutler samples as a thin film on the sand grains as well as in the cement argues that the red staining (and thus magnetization) took place before diagenesis was complete.

These considerations and the general agreement with other Permian results from North America (for a summary see *McMahon and Strangway* [1968*a*, *b*]) favor a Permian age for the stable magnetization.

KIAMAN MAGNETIC DIVISION

The time span represented by the Cutler formation is not known. If it is restricted to the middle and upper Wolfcampian and lower Leonardian [*Baars*, 1962; *Baars and Seager*, 1970], then the average rate of deposition is about 1 cm/1000 years. The spacing between the samples used in this study is about 0.6 meters, which is equivalent to 60,000 years. If the Cutler formation covers a shorter time span, the time

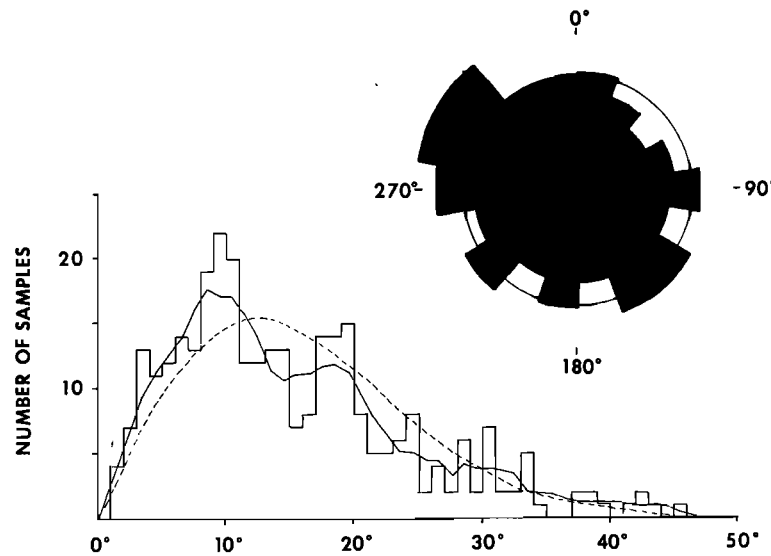


Fig. 11. Comparison of observed and expected frequencies of azimuthal and angular deviations from the mean. In the azimuthal diagram the circle corresponds to the expected frequency of 17.6. The histogram in the lower figure is based on 1° intervals, the continuous curve is the gliding average over 5°, and the dashed curve represents the expected distribution. Number of samples is 321.

TABLE 1. Summary of Paleomagnetic Data after Thermal Demagnetization to 520 ± 20°C

Sample	Number of Samples	Precision Parameter k	α_{95}*	Lat., °N	Long., °E
All Cutler	321	20.8	1.8	44.4	116.2
1	8	23.7	11.6	50.5	103.4
2	6	13.1	19.2	48.6	109.9
3	7	18.3	14.5	48.1	115.9
4	9	38.8	8.4	43.5	113.3
5	5	34.9	13.1	45.5	107.7
6	6	10.2	22.0	39.9	116.8
7	5	270.0	4.7	51.1	111.2
8	9	72.5	6.1	47.4	117.0
9	11	29.8	8.5	45.9	118.4
10	10	35.2	8.3	46.5	113.6
11	13	19.2	9.7	43.8	116.8
12	12	16.8	10.9	45.0	115.9
13	10	33.2	8.5	35.1	122.4
14	9	13.9	14.3	45.7	113.8
15	9	38.0	8.5	44.0	111.9
16	10	24.9	9.9	41.6	111.8
17	8	12.9	16.0	47.9	114.6
18	11	174.7	3.5	45.3	117.0
19	10	13.8	13.5	36.2	120.5
20	5	29.0	14.4	35.8	114.2
21	14	20.0	9.1	40.8	118.1
22	21	17.8	7.9	47.3	122.5
23	13	14.8	11.1	47.9	110.9
24	12	25.9	8.7	46.3	116.6
25	8	24.3	11.5	34.1	115.6
26	20	35.8	5.5	49.5	114.9
27	13	14.4	11.3	45.0	120.6
28	23	48.6	4.4	43.8	119.4
29	8	59.5	7.2	41.7	111.8
30	9	10.3	16.8	38.2	121.1
31	7	51.2	8.5	43.8	115.0
Mean	31	105.2	2.5	44.1	115.4
All Elephant Canyon	28	27.2	5.3	43.6	119.6
1	16	22.1	8.0	44.8	118.8
2	6	26.5	13.3	38.8	125.2
3	6	71.3	8.0	45.2	115.5

*Semi-angle of 95% circle of confidence.
All poles have been transferred to the northern hemisphere. Individual samples for Cutler and Elephant Canyon were taken at 6-meter intervals in descending order.

interval between samples becomes smaller. We therefore may state that in our section no normal interval exists with a duration longer than 60,000 years. This is in agreement with the results of *McMahon and Strangway* [1968*a*, *b*].

Three normal events have been reported in the Upper Permian [*Guicherit*, 1964; *Petrova*, 1965; *Dachroth*, 1969]. They have not been found in the present study in spite of dense stratigraphic sampling. Their absence supports the conclusion that the Cutler formation at this locality does not extend higher than the Leonardian stage.

At the base of the Permian a normal event has been found by *Graham* [1955] in the Supai formation and by *Helsley* [1965] in the Dunkard group. The two authors possibly recorded the same event and it has been suggested that it should be named the 'J. W. Graham event' [*Burek*, 1970]. Since these samples were not adequately demagnetized, it was considered worthwhile to subject them to stepwise thermal demagnetization. Two of the three samples of the Dunkard group with a normal direction (Figure 3*a* in *Helsley* [1965]) proved to be magnetically stable. The samples from the Supai

formation (Figure 7, site 11*A* and 11*B* in *Graham* [1955]) were made available by J. W. Graham but proved to be unstable and it is, therefore, not clear whether a normal event exists in the Oak Creek section.

The J. W. Graham event was not observed in the Cutler formation or in the upper part of the Elephant Canyon formation. It is unlikely that it could have been missed in our densely sampled section. However, it could be present in the interval between 25 and 50 meters above the base of the Cutler or in the lower part of the Elephant Canyon formation.

Pole Positions

Before a meaningful pole position can be calculated, we must ascertain whether the directions obey a Fisher distribution [*Fisher*, 1953; *Watson and Irving*, 1957]. Figure 11 compares the observed and expected frequencies for the azimuthal and angular deviations from the mean. Although the agreement is not perfect, the discrepancies are not statistically significant. The chi-square test yields P ($\chi^2 \leq 13.6$) $\simeq 0.40$ for the azimuthal distribution and P ($\chi^2 \leq 17.6$) $\simeq$ 0.25 for the angular distribution; the latter value is based on the gliding average over 5°. To further test the acceptability of the demagnetized results, the 321 observations were subdivided into 5 groups according to the angular distance between their NRM directions and the thermally demagnetized direction. Within their 95% circle of confidence all 5 groups yield the same value as the total mean; this result proves that the thermal demagnetiation was successful in removing the secondary component.

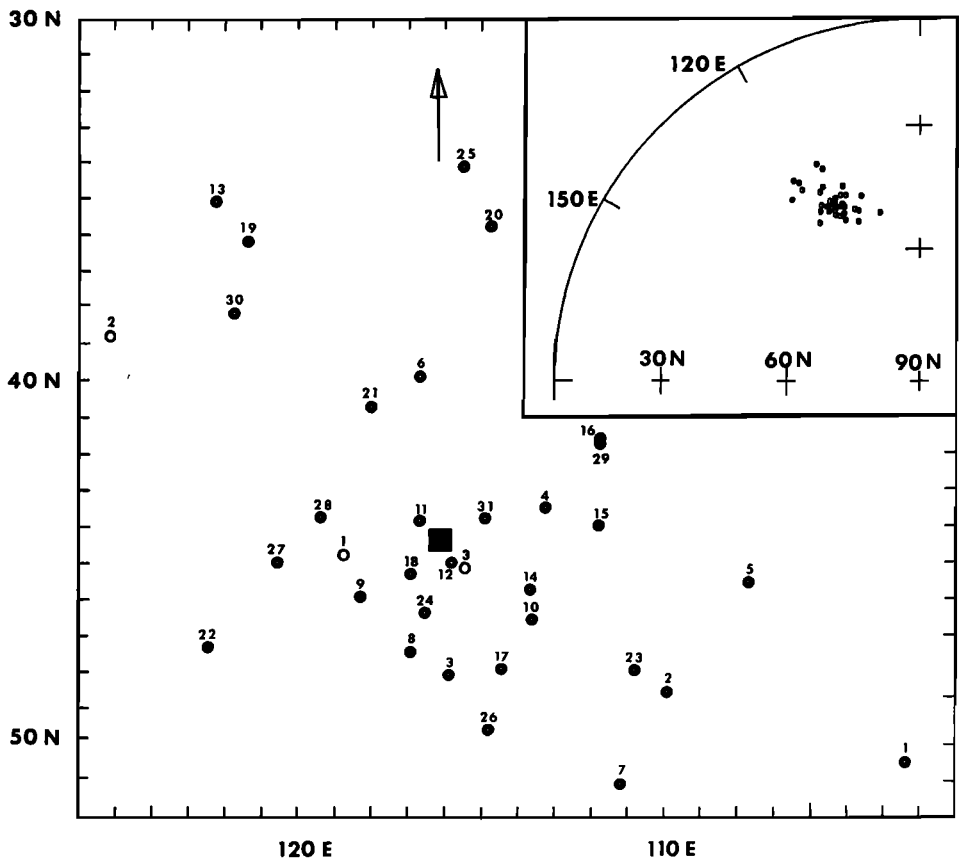


Fig. 12. Pole positions of 6-meter intervals. Solid circles represent the Cutler formation; open circles represent the Elephant Canyon formation. Numbers are assigned in descending order. Square is the mean of all Cutler data.

Table 1 summarizes the paleomagnetic data, together with some statistical parameters [*Fisher*, 1953; *Irving*, 1964]. The pole position derived from the thermally demagnetized Cutler samples is at 44.4°S, 63.8°W; the equivalent pole in the northern hemisphere is at 44.4°N, 116.2°E. Because of the large number of samples ($n = 321$), the 95% circle of confidence becomes very small, namely 1.8°. The samples from the Elephant Canyon formation yield a pole position at 43.6°N, 119.6°E. The circle of confidence is 5.3° and includes the Cutler pole. The two poles are therefore nearly identical.

The Cutler data seem well suited for a determination of the variation in pole positions one may expect in a single sedimentary unit if only a limited number of outcrops are sampled. The section has arbitrarily been subdivided into 6-meter segments, which may represent times of the order of 1 m.y. The statistical parameters for these 31 intervals are included in Table 1 and the pole positions are shown in Figure 12. The numbers in Figure 12 correspond to the 6-meter intervals in descending order; the arrow indicates the direction in which the secondary (present day) component would displace the pole. The figure shows that the secondary component has successfully been removed, since there is no streaking in the direction of the present day field. Individual means differ by as much as 15°, even for adjacent intervals. Clearly, a small number of outcrops can give rather erroneous results and a large stratigraphic coverage is necessary for obtaining a representative mean. The mean pole position for the Cutler formation remains the same whether unit weight is assigned to each sample or to the 6-meter means; however, the α_{95} is increased slightly as one would expect as a result of the reduction in the number of independent samples being averaged in the mean.

The agreement with published paleomagnetic poles for North America is good (for a summary see *McMahon and Strangway* [1968*a*, *b*]). The results from the Cutler formation by *Farrel and May* [1969] and *Helsley* [1971] are the same within the 95% confidence limit. *Helsley* [1971] remarked that a change in pole position of approximately 20° occurred between the Lower Permian and Lower Triassic. The present data seem to confine this change to the Upper Permian only. Alternately, one could argue that the 5° change in declination that was obtained by fitting a straight line through the data points would be significant if more data were available. If, for instance, a straight line is fitted through the gliding average over 5 samples, this change becomes significant, but in doing so each sample is counted 5 times. This indicates that a much larger population than ours is necessary in order to ensure statistical significance of small trends. However, the suggested change is toward the Lower Triassic direction and may, therefore, be real. Assuming that the 20° change in pole position during the Permian was uniform in time, our data imply that the Cutler formation would be restricted to the lowest fourth of the Permian (Wolfcampian); this theory is completely consistent with the regional stratigraphic framework.

Acknowledgments. The authors thank the late Dr. J. W. Graham for permission to demagnetize his samples from the Supai formation. The paper greatly benefited from discussions with Dr. A. L. Hales and the late Dr. H. Porath, and a constructive review of the manuscript by Dr. E. M. Shoemaker was particularly valuable.

This work was supported by the National Science Foundation under grants GA 1304 and GA 15999. Final editing was done at The Lunar Science Institute under the joint support of the Universities Space Research Association and the NASA Manned Spacecraft Center under contract NSR 09-051-001.

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(Received April 5, 1971;
revised December 9, 1971.)